WIFI AP Mode Experiment

1. WIFI AP Mode Overview

WiFi AP (Access Point, hotspot mode) lets the ESP32-S3 turn itself into a **wireless router/hotspot**; other devices (phones, computers, another ESP32) can search for and connect to this hotspot to achieve LAN communication.

AP mode has **irreplaceable application value** in the IoT field. First, it is the **foundation of provisioning technology** — when a device is first deployed and no WiFi network is available, the device first runs in AP mode, the user connects their phone to the hotspot and sends the router password to the device through a web page or App, and the device then switches to STA mode to go online (this is the first step of the SmartConfig / Web provisioning process). Second, in **no-router scenarios** (field work, site inspections, temporary exhibitions), AP mode lets users directly connect to devices point-to-point for parameter configuration, status viewing, and firmware upgrades without depending on any network infrastructure. In addition, AP mode is also widely used in **LAN multi-user collaboration** scenarios — multiple phones or computers connect to the same ESP32-S3 hotspot for local chat, file sharing, multiplayer games, and other LAN applications.

This experiment configures the ESP32-S3 in AP mode; a phone or computer can view the connection status after connecting.

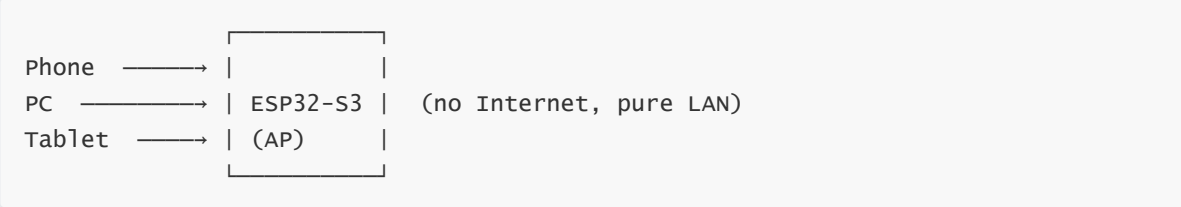
Typical application scenarios:

Application Type	Example
Device provisioning	Phone connects to ESP32 hotspot -> enter home WiFi on web page -> ESP32 switches to STA
Offline LAN	Multiple ESP32s form an ad-hoc network (no router needed)
Debugging	Connect directly for debugging when there is no WiFi in the field
File transfer	Phone connects to ESP32 hotspot and uploads/downloads files

> In AP mode the ESP32-S3 supports about 4~8 devices connecting simultaneously at most.

2. Core Concepts

2.1 How AP Mode Works



The ESP32-S3 creates its own hotspot and broadcasts the SSID; other devices scan and connect. The ESP32-S3 acts as a DHCP server to assign IPs to clients. In AP mode, the ESP32-S3's default IP is usually `192.168.4.1`.

2.2 Configuration Parameters

Parameter	Description	Value Used in This Experiment
AP_SSID	Hotspot name	`"ESP32_AP"`
AP_PASSWORD	Hotspot password	`"12345678"` (at least 8 characters)
authmode	Encryption method	`WPA_WPA2_PSK`
Max connections	Number of simultaneously connected devices	4~8 (depends on firmware configuration)

2.3 Core API

```
import network

ap = network.WLAN(network.AP_IF)           # create AP interface
ap.active(True)                             # activate
ap.config(essid="ESP32-AP",                # set SSID + password
          password="12345678",
          authmode=network.AUTH_WPA_WPA2_PSK)
ip, _, _, _ = ap.ifconfig()                # get AP's own IP
ap.status('stations')                       # get connected clients
ap.active(False)                            # deactivate
```

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3. Hardware Dependencies

The ESP32-S3 has a built-in 2.4 GHz WiFi module, **no external hardware needed**.

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4. Project Architecture and Flow

```
Script execution (single-file .py)
  |
  |- create AP interface -> set SSID/password/encryption
  |- wait for AP activation -> print IP
  |
  |- while True:
       |- poll the list of connected clients
       |- detect new connection -> print MAC address
       |- detect disconnect -> print MAC address
       |- sleep(1s)
```

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5. Source Code Explained

Full source code: `Source Code -> MicroPython -> 4.Network Protocols -> WiFi_AP -> WiFi_AP.py`

5.1 Configuration and Activation

`AP_IF` interface -> `active(True)` -> `config(ssid, password, authmode)` sets the hotspot parameters -> wait for activation to complete.

```
AP_SSID     = 'ESP32_AP'                # AP SSID
AP_PASSWORD = '12345678'                    # AP password (min 8 chars)

def main():
    ap = network.WLAN(network.AP_IF)        # create AP interface
    ap.active(True)                          # activate
    ap.config(essid=AP_SSID, password=AP_PASSWORD,
              authmode=network.AUTH_WPA_WPA2_PSK)  # WPA2 encryption

    while not ap.active():
        time.sleep_ms(100)                   # wait for activation

    ip, _, _, _ = ap.ifconfig()
```

```
    print('WiFi AP ready')
    print('  SSID     :', AP_SSID)
    print('  IP       :', ip)
```

5.2 Client Connection Monitoring

`status('stations')` gets the MAC list of connected devices; comparing two consecutive snapshots implements connect/disconnect detection.

```
def mac_str(mac):
    """bytearray -> readable string"""
    return '%02x:%02x:%02x:%02x:%02x:%02x' % (mac[0], mac[1], mac[2],
                                              mac[3], mac[4], mac[5])

prev = []
while True:
    cur = [t[0] for t in ap.status('stations')]  # current clients

    for mac in cur:                              # new connections
        if mac not in prev:
            print('Connected   :', mac_str(mac))

    for mac in prev:                             # disconnected
        if mac not in cur:
            print('Disconnected:', mac_str(mac))

    prev = cur
    time.sleep(1)                                # 1s polling interval
```

> `status('stations')` returns a list of (MAC, ...) tuples of connected devices; connect/disconnect detection is implemented by comparing two consecutive snapshots.

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6. Operating Procedures

6.1 Flashing and Running

Thonny IDE steps: See `MicroPython -> 1. Before Development -> 3.Thonny IDE`.

Thonny IDE flashing steps: See the `MicroPython -> 1. Before Development -> 3.Thonny IDE`

1. Open `WiFi_AP.py` in Thonny IDE and click Run
2. Open the WiFi settings on your phone and search for `ESP32_AP`
3. Enter the password `12345678` to connect

6.2 Expected Results

After the ESP32-S3 creates the hotspot, the phone can find and connect to it. The serial port prints client connect/disconnect logs.

6.3 Serial Output Example

```
WiFi AP ready
  SSID     : ESP32_AP
  IP       : 192.168.4.1
Connected   : aa:bb:cc:dd:ee:ff
Disconnected: aa:bb:cc:dd:ee:ff
```

Hotspot is on, phone connects successfully

```
WiFi AP ready
  SSID     : ESP32_AP
  IP       : 192.168.4.1
Connected   : e6:d1:24:85:df:49
```

Phone disconnects

```
Disconnected: e6:d1:24:85:df:49
```

6.4 Verification Methods

Operation	Expected Result
Phone searches for WiFi	Sees `ESP32_AP`
Phone enters the password and connects	Serial prints `Connected: <MAC>`
Phone disconnects from WiFi	Serial prints `Disconnected: <MAC>`

7. Common Problems

Symptom	Cause	Solution
Phone cannot find the hotspot	AP not activated or 5 GHz band incompatible	The ESP32-S3 only supports 2.4 GHz; confirm the phone is searching in the 2.4G band
Prompts "wrong password" when connecting	Password shorter than 8 characters	WPA2 requires a password of at least 8 characters
Cannot access the Internet after connecting	AP mode has no Internet	AP mode only provides a LAN and does not forward Internet traffic